In [3]:

```
!pip install ultralytics opencv-python matplotlib
```

```
Requirement already satisfied: ultralytics in /usr/local/lib/python3.13/dist-packages (8.4.163)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.13/dist-packages (5.0.0.93)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)
Requirement already satisfied: cloudpickle>=3.1.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(3.1.2)
Requirement already satisfied: filelock>=3.16.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(3.32.5)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(2.1.3)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(2.32.4)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(2.11.0+cpu)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(0.26.0+cpu)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
(1.35.2)
Requirement already satisfied: nvidia-ml-py>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
 (13.615.71)
Requirement already satisfied: ultralytics-thop>=2.1.6 in /usr/local/lib/python3.13/dist-packages (from
ultralytics) (2.1.6)
Requirement already satisfied: ultralytics-platform>=0.1.45 in /usr/local/lib/python3.13/dist-packages (from
ultralytics) (0.1.65)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib)
(4.64.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib)
(1.5.1)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib)
(26.3)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib)
(3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib)
(2.9.0.post0)
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.13/dist-packages (from
polars>=0.20.0->ultralytics) (1.35.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.7-
>matplotlib) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from
requests>=2.23.0->ultralytics) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0-
>ultralytics) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from
requests>=2.23.0->ultralytics) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from
requests>=2.23.0->ultralytics) (2026.7.22)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from
torch>=1.8.0->ultralytics) (4.16.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0-
>ultralytics) (80.10.2)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0-
>ultralytics) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0-
>ultralytics) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics)
 (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0-
>ultralytics) (2025.12.0)
Requirement already satisfied: httpx<1,>=0.28 in /usr/local/lib/python3.13/dist-packages (from ultralytics-
platform>=0.1.45->ultralytics) (0.28.1)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28-
>ultralytics-platform>=0.1.45->ultralytics) (4.14.2)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28-
>ultralytics-platform>=0.1.45->ultralytics) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,
>=0.28->ultralytics-platform>=0.1.45->ultralytics) (0.16.0)
```

```
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)
```

In [4]:

```python
import cv2
import matplotlib.pyplot as plt
from ultralytics import YOLO
```

In [5]:

```python
# Load the pre-trained YOLOv8 model
model = YOLO("yolov8n.pt")

print("YOLOv8 model loaded successfully!")
```

```
YOLOv8 model loaded successfully!
```

In [6]:

```python
# Download a sample image
!wget -q https://ultralytics.com/images/bus.jpg -O bus.jpg

# Load the image
image = cv2.imread("bus.jpg")

# Convert BGR to RGB
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Display the input image
plt.figure(figsize=(10, 6))
plt.imshow(image)
plt.title("Input Image")
plt.axis("off")
plt.show()
```



In [7]:

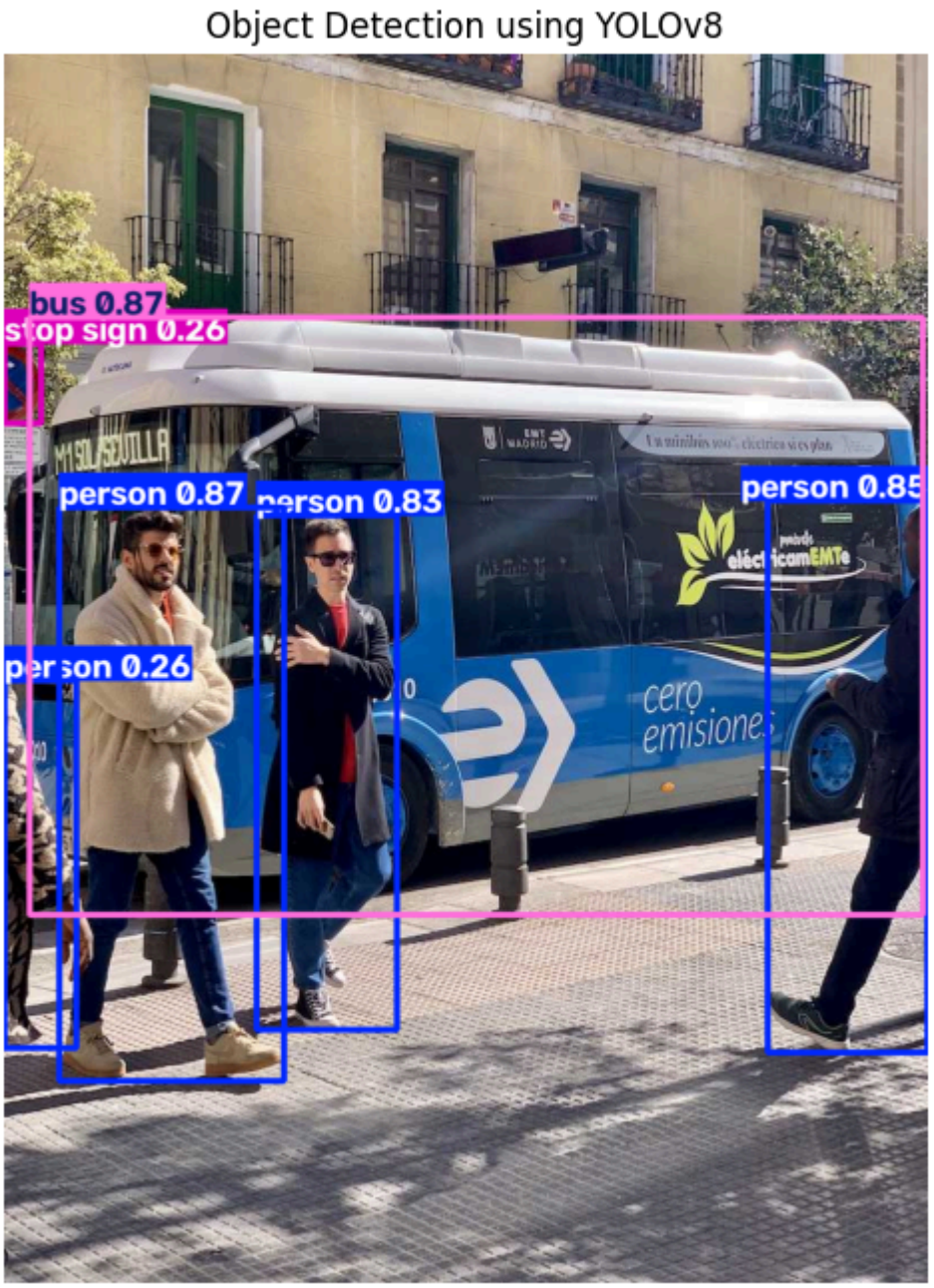
```python
# Perform object detection
results = model("bus.jpg")

# Draw bounding boxes and labels
output = results[0].plot()
```

```
# Convert BGR to RGB
output = cv2.cvtColor(output, cv2.COLOR_BGR2RGB)

# Display the detected objects
plt.figure(figsize=(12, 8))
plt.imshow(output)
plt.title("Object Detection using YOLOv8")
plt.axis("off")
plt.show()
```

```
image 1/1 /content/bus.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 409.6ms
Speed: 33.1ms preprocess, 409.6ms inference, 36.0ms postprocess per image at shape (1, 3, 640, 480)
```



In [8]:

```
# Display detected objects and confidence scores
for box in results[0].boxes:
    class_id = int(box.cls[0])
    confidence = float(box.conf[0])
    label = model.names[class_id]

    print(f"Object: {label}, Confidence: {confidence:.2f}")
```

```
Object: bus, Confidence: 0.87
Object: person, Confidence: 0.87
Object: person, Confidence: 0.85
Object: person, Confidence: 0.83
Object: person, Confidence: 0.26
Object: stop sign, Confidence: 0.26
```

Experiment Name: Object Detection and Recognition Using Deep Learning Models on Standard Image Datasets

Aim: To implement a deep learning-based object detection and recognition system using a pre-trained YOLOv8 model to identify and locate multiple objects in an image.

Theory: Object detection is a computer vision technique used to identify and locate objects in an image. YOLOv8 is a deep learning model that detects multiple objects and provides bounding boxes, class labels, and confidence scores.

Procedure:

Install and import the required libraries.

Load the pre-trained YOLOv8 model.

Load a sample image.

Perform object detection on the image.

Display the detected objects with bounding boxes and labels.

Print the object names and confidence scores.

Result: The object detection system was successfully implemented using the pre-trained YOLOv8 model. The model identified objects in the input image and displayed their bounding boxes, class labels, and confidence scores.